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# **Gacha system in free-to-play games**

## **BACHELOR THESIS**

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## **Abstract**

This bachelor thesis deals with gacha mechanism, compares three different gacha systems in free-to-play games, and takes a deeper look into the Arknights gacha system whose simulation this work aims to create. Thus the main aim of this thesis is to create a complex simulation in the programming language Python and analyze its predictions and results. This thesis also describes the rules and common design practices used in games implementing this system and puts the simulation of one specific system into the context of other existing implementations, the history of gacha and similar random rewards mechanisms, and a probability theory. In the chapter dedicated to probability theory, this thesis explains related terms such as stochastic process, stochastic simulation, Markov process, Markov property, and Markov Chain Monte Carlo method, which is a method in a modified form used in the design and execution of the simulation. The results show how essential the implementation of pity is for the whole system. The results also state the prediction that the average resources needed to acquire the highest rarity reward are 48,000, meaning approximately four months' worth of standard game rewards or premium currency worth of 200\$. This thesis can be used both by the players who want to gain a better understanding of the gacha system, as well as the developers of games implementing the said system. It also puts current knowledge regarding legislation, regulations, and the psychological aspect of this phenomenon into a new perspective.

## **Keywords**

simulation, probability, prediction, game of chance, Python

## Abstrakt

Tato bakalářská práce se zabývá gacha systémy, porovnává tři různé systémy těchto volně dostupných her, a zejména se zaměřuje na gacha systém hry Arknights, jehož simulaci si klade za cíl vytvořit. Hlavním cílem práce tedy je vytvoření komplexní simulace v programovacím jazyce Python, a následná analýza predikcí a výsledků. Tato práce popisuje také pravidla a běžné návrhové praktiky užívané ve hrách implementujících gacha systém a klade vytvořenou simulaci do kontextu ostatních implementací, historie gacha a podobných mechanismů náhodných odměn, a teorie pravděpodobnosti. V kapitole věnované teorii pravděpodobnosti práce uvede související pojmy jako je náhodný proces, stochastická simulace, Markovův proces a Monte Carlo pomocí Markovova řetězce, jehož metoda je v upravené podobě použita v simulaci. Výsledky ukazují míru vlivu pitty na celý systém. Dále výsledky predikují průměrné potřebné zdroje pro dosažení nejvyšší odměny, ty představují 48.000, což představuje asi čtyři měsíce běžných herních odměn nebo 200\$ v placené měně. Tato práce může být přínosná jak pro hráče, kteří mohou lépe porozumět fungování gacha systému, tak pro vývojáře her implementujících tento systém. Dále práce uvádí současný stav poznání zejména v oblasti legislativy, regulací a z pohledu psychologie na danou problematiku do nové perspektivy.

## Klíčová slova

simulace, pravděpodobnost, predikce, hazard, Python

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# Introduction

The popularity of games implementing the gacha system is growing beyond Asian culture. These games, usually available as mobile, free-to-play applications, are generally known as gacha games. Even studios that created multiple titles games implementing the standard payment model (one-time payment) are trying to implement the gacha system in their new titles (*Limbus Company*, 2023). It is not an easy task as running a gacha game does not require only decent gameplay, well implemented client-server model, a good story and atmosphere, but also a fitting gacha system connected to the other parts of the game.

Current research is focused chiefly rather on the psychological (Shibuya et al., 2015; Nielsen and Grabarczyk, 2019; Klézl and Kelly, 2023), sociological (B. Britt and R. Britt, 2021; Woods, 2022) and regulatory (Koeder and Tanaka, 2017; Carvalho, 2021) aspects of this phenomenon. Only in recent months papers focused on the gacha game model, its design, and optimization (Chen and Fang, 2023) have been published. Most of the mentioned sources do not focus only on one area but connect named aspects and sometimes consider the perspective of economics or design in gacha games. Even with the recent research progress, the gacha system, especially its mechanism, needs to be looked into more closely. Not only could work focused on an analysis of the gacha system and especially the rolling/pulling mechanisms, provide additional context and contribute to both existing and following work related to gacha games and their system, but it could also provide information for both the players and the developers of gacha games. Above mentioned, alongside the author's interest in this topic, are the reasons why this paper was written.

This work aims to create a complex simulation of the gacha system of the free-to-play game Arknights. Without further context, this simulation would not provide the necessary data to achieve the intended goal mentioned in the last paragraph.

The methodology of this work is specified for each use case. Methods used in the chapter 1 consist of a literature review alongside with author's observation, analysis, and comparison of various online sources, including the official websites of Arknights (*Arknights*, 2020), Faith/Grand Order (*Fate/Grand Order (English)*, 2015), and Genshin Impact (*Genshin Impact*, 2020). The main reasons these three video games were selected for closer analysis and comparison are mentioned in section 1.4. The named methods were predominant in the first chapter, where the gacha model is introduced. A literature review was the primary method in the second chapter. The second chapter sets the gacha system in the context of statistics and probability theory. It emphasizes the connection of the theoretical concepts to gacha mechanisms, especially to the studied system in the game Arknights. Observation and analysis are methods persistent in chapter 3. In this chapter, multiple simulations in specific scenarios are executed, and their output is analyzed. The Markov Chain Monte Carlo method is generally followed in these random experiments. This method and the changes applied to it are specified in the chapter 2 of this bachelor thesis.

# 1. Gacha model

Gacha is a widely used game model, especially in Asian culture. Japan, Korea, and China seem to be implementing the gacha system in free-to-play video games the most. Still, it is not exclusive to these regions, thanks to globalization and the development and introduction of similar random reward mechanisms in games. A typical example of these mechanisms in games can be the process of opening new card packs or loot-boxes. Just like loot boxes or card packs, gacha is an implementation of random procedures used to select and deliver rewards in video games (Nielsen and Grabarczyk, 2019).

Therefore we will first look at all the random rewards mechanisms and later specify aspects exclusive to the gacha system. First, in section 1.2, we will take a brief look into the history of random reward mechanisms. Then, in section 1.1 author defines relevant terms. Later, in section 1.3, we will introduce the free-to-play model and explain its connection to gacha systems. Last but not least, we will describe the essential elements of the gacha model in section 1.4. Mentioned section draws upon literature and review of online sources and the author's observation and analysis of game-play in the three games selected for comparison.

## 1.1 Definition

Video games that implement the gacha mechanism are generally referred to as gacha games. The term gacha originated in Japan and has roots in the word Gachapon. Gachapon is a trademark of Bandai Co., Ltd. (Bandai Co., Ltd., 2023) and a name for vending machines that dispense plastic capsules containing toys. Shibuya, Teramoto, and Shoun describe the similarity between gacha and Gachapon as follows: *Players can also use gacha, which are similar in screen appearance to vending machines that dispense children's toys, and lucky players can win valuable gaming items this way. Gacha is onomatopoeia that imitates the sound of a crank on a toy vending machine. Gacha can be played for free, however, extremely rare and/or valuable gaming items can also be obtained through monetary purchases of online gacha products* (Shibuya et al., 2015, p. 3).

Gacha and related terms are not quite well defined for academic usage. Based on studied literature and observation of real gacha systems author specifies the following definitions for further usage in this bachelor thesis:

- **Gacha** is generally used as a synonym to gacha game, but depending on the given context can refer to the gacha mechanism or the whole system. It also sometimes refers to the vending machine type of gacha. Therefore, this work will avoid separate usage of this term.
- **Gacha mechanism** is a video game mechanism in which players can spend real money or virtual currency to acquire randomized virtual items or characters. From a statistical



point of view, we can see this mechanism as a variation of a random/stochastic process, more about this process in chapter 2. It consists of algorithms and probabilities that work together to produce randomized results.

- **Gacha system** is the implementation of a gacha mechanic in a video game or other software. It consists of multiple elements and their interactions, see section 1.4 for more information about the elements.
- **Gacha model** refers to guidelines, principles, and methods followed in video games implementing the gacha system.
- **Gacha game** is a video game that fulfills the gacha model.

## 1.2 History of gacha and similar mechanisms

This section introduces the history of gacha and similar random reward mechanisms. Recent trends, implementation of similar mechanisms in video games and analog media in the past, and randomness in everyday life, are discussed in this section. It explains the popularity of the gacha model and helps us improve our understanding of the gacha mechanism.

In a paper from 2019, Nielsen and Grabarczyk discuss random reward mechanisms, namely loot-boxes, in games (Nielsen and Grabarczyk, 2019). Gacha games, by all means, not only apply random reward mechanisms but also glorify the act of randomness itself. This glorification of randomness can be perceived as a relatively new trend in video games (Nielsen and Grabarczyk, 2019).

Various forms of random reward mechanisms are common not only in video games but in our everyday life as well. We can observe this mechanism in any video game where killing a certain enemy or other action performed by a player triggers a random process that may result in awarding the player with a new object, skill, or different reward. We can see the main components of this mechanism in the following figure 1.1. These random reward mechanisms were used almost from the beginning of the medium (Nielsen and Grabarczyk, 2019) and even before we could encounter these mechanisms in analog games. The throw of a dice, the draw of a card, and collectible cards that one buys without knowing what the pack contains are all examples of random reward mechanisms. Some collectible cards even introduced the concept of rarity, which is an essential part of the gacha mechanism.

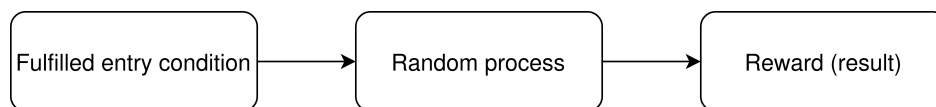


Figure 1.1: Three main elements of any random reward mechanism (by author)

It is important to note that these mechanisms are not exclusive to gaming since we encounter randomness in our everyday life. The current cryptography could be a good example of randomness in use, but our focus will be more on the usage of randomness related to the gacha system in the core idea. Chocolate eggs that contain a random toy, marketing programs

in retail that reward customers with a random item, such as a collectible sticker or toy, after spending a set amount of money. Even the paper by Nielsen and Grabarczyk mentions the similarity of random reward mechanisms to some marketing practices in the past, like collectible pictures attached to cigarettes back in the 19th century (Nielsen and Grabarczyk, 2019). It may be because both these marketing practices and the gacha system aim to increase the income of the companies implementing them.

### 1.3 Free-to-play games

The gacha game model does not consist of the gacha mechanism alone. It can be considered a subcategory of the freemium business model. We can observe features in other market models, as well as in video games. *In software markets, this strategy implies the development of a full-featured version of software and another low-quality version by disabling a subset of features* (Sánchez-Cartas, 2022, p. 2).

The freemium model in video games is widely recognized as free-to-play (F2P). The essential component or feature of F2P is that players acquire the game for free and then pay for in-game items (Sánchez-Cartas, 2022). In the case of the gacha model, the players do not buy the desired item, character, or other reward directly. Instead, the players exchange money for virtual currency that can serve various purposes. The hard in-game currency can always be used, after possible transformations, to pay the fee necessary for players to partake in the gacha mechanism (*Fate/Grand Order (English)*, 2015; *Genshin Impact*, 2020; *Arknights*, 2020).

### 1.4 Essential elements of the gacha model

At the outset, it is important to state that the gacha mechanism is the key element but not the game itself (Koeder and Tanaka, 2017). Looking at figure 1.2, we can see how essential Gacha is for the economic model and the whole game ecosystem. This thesis mainly focuses on the Gacha mechanism but will also introduce the essential parts of any gacha system.

The description of the gacha model is built upon a literature review, analysis of online sources, and in-game content in mainly three different gacha games. The games *Arknights*, *Fate/Grand Order*, and *Genshin Impact* were selected for analysis and comparison because of their difference in the implementation of gacha system (*Arknights*, 2020; *Fate/Grand Order (English)*, 2015; *Genshin Impact*, 2020), for the reasons of the authors' familiarity with these systems, and the popularity of *Genshin Impact*. The *Fate/Grand Order*, released in 2015 (*Fate/Grand Order (English)*, 2015), can also be seen as one of the first games implementing the gacha system as we know it today.

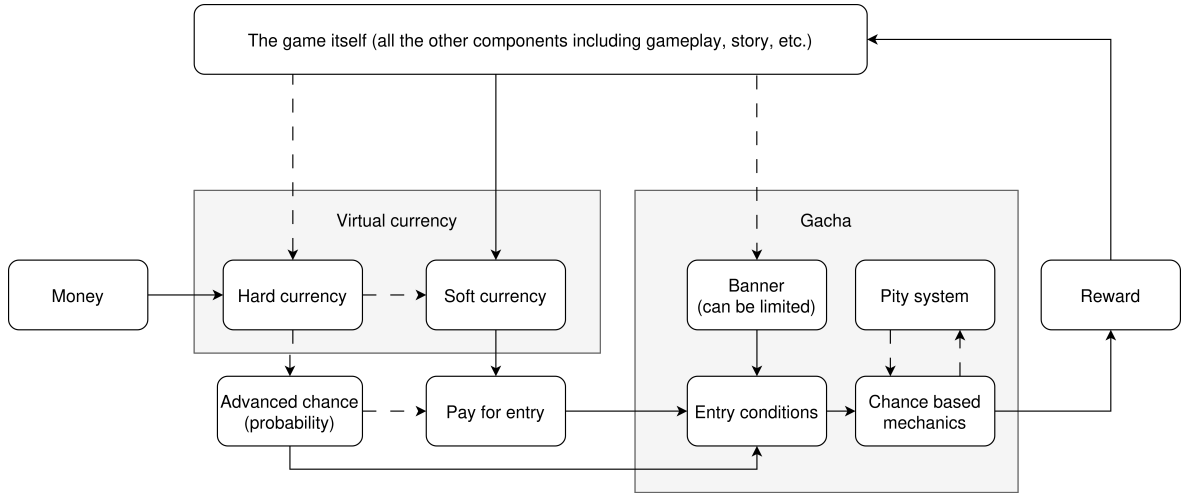


Figure 1.2: Key elements of gacha system based on observation, comparison of different gacha systems, previous research, and game analysis. Partly taken from a chart displaying the outline of Gacha elements in Japanese mobile F2P games (Koeder and Tanaka, 2017) (by author)

Looking at figure 1.2, we can observe the main elements of gacha system and its economic model. The purpose of this chart is to clarify the purpose and interactions of the critical elements in gacha system. The solid lines describe the necessary links and interactions in the system. In contrast, the dashed lines describe optional links, which depend on the specific gacha system implementation and can vary depending on the game or specific banner.

The link between the game itself and soft currency symbolizes that players earn resources to participate in the gacha mechanism by playing the game. Some games make it possible to earn premium currency this way as well. For example, *Arknights* has this functionality and rewards every first clear of any stage with hard currency (*Arknights*, 2020). We do not consider that virtual currency may serve multiple purposes in games and focus merely on the part connected to the gacha mechanism. There may also be other kinds of currency used in order to access the gacha mechanism, such as vouchers. There can also be different kinds of rolling currency, depending on the type of banner, like in *Genshin Impact* (*Genshin Impact*, 2020).

### 1.4.1 Rolling

Rolling, sometimes also referred to as pulling, is how the whole use of gacha mechanism is perceived and called by the players. The term roll describes this most accurately, but this work may also refer to its instances as pulls. One roll is one activation or one run through a random process. In figure 1.2, we see that entry conditions have to be met in order to start this mechanism. Entry conditions include a decision to engage with the mechanism, the price for entry being paid, and a set of conditions and custom values (probability space) that depends on a specific banner. Once the conditions are met, the random process, part of the

gacha mechanism called one roll, activates. The system stores values related to its current state, and the outcome of this process is always a reward delivered to the player. Gacha games also usually have the option of 10 rolls in place of one. Because of this, the system applying 10 rolls can include additional conditions like changes in the price or probability space. An example of this can be the one additional free roll for a player in Faith/Grand Order (*Fate/Grand Order (English)*, 2015). Arknights, on the other hand, does not apply any changes in the system (*Arknights*, 2020).

### 1.4.2 Pity system

Pity system, also called pity counter or pity, is an optional part of any gacha system. Connected to the primary mechanism, it changes the initial probabilities depending on the results of previous draws, the number of rolls, or other conditions and their combination. Its main objective is to make the random process less random by guaranteeing a high rarity, the highest rarity, or a specific reward in a given number of rolls. Correct implementation of this system reduces the level of uncertainty.

Because the simulation created in the third chapter includes pity system, to understand how exactly a pity system can work will be best to look at an implementation of pity system in three games implementing the gacha system.

We need to introduce the terms soft pity and hard pity. Soft pity increases the chances of getting a target reward or reward of target rarity, it usually activates once, and the probability keeps growing in every roll. On the other hand, hard pity forces the gacha system to return the target reward or high rarity reward in a given number of rolls. Arknights (*Arknights*, 2020) and Genshin Impact (*Genshin Impact*, 2020) use a combination of both. Faith/Grand Order did not use to have a pity system but now implements hard pity (*Fate/Grand Order (English)*, 2015). The table 1.1 shows differences in the implementation of the pity system in three different gacha games (*Arknights*, 2020), (*Fate/Grand Order (English)*, 2015), (*Genshin Impact*, 2020).

Table 1.1: Pity system in gacha games (*Arknights*, 2020; *Fate/Grand Order (English)*, 2015; *Genshin Impact*, 2020) compared (by author)

| Name             | Pitty carries over<br>between banners | Rolls to activate: |           | Limited<br>usage |
|------------------|---------------------------------------|--------------------|-----------|------------------|
|                  |                                       | Soft pity          | Hard pity |                  |
| Arknights        | <i>Yes</i>                            | 50                 | 100       | <i>No</i>        |
|                  | <i>No</i>                             | -                  | 10        | <i>Yes</i>       |
| Fate/Grand Order | <i>No</i>                             | -                  | 330       | <i>Yes</i>       |
| Genshin Impact   | <i>Yes</i>                            | 75                 | 90        | <i>No</i>        |
|                  | <i>Yes</i>                            | -                  | 10        | <i>No</i>        |

*Note:* Some games implement multiple pity mechanisms, this table considers standard banners

Both kinds of pity reset after the target reward or reward of target rarity is acquired. Sometimes the pity system has limited usage, meaning the system stays disabled for the specific player and banner. On the other hand, pity can be carried over between banners, but this can be limited to certain types of banners.

### 1.4.3 Login and task rewards

Other elements of the gacha model are login and task rewards. These rewards usually include rewards for daily, weekly, monthly, or seasonal quests. Players, upon completion of these tasks, receive a variety of in-game items and resources. The rewards depend on the specific game but almost always include virtual currency. From the game's creators' point of view, the primary purpose of login and task rewards is to keep players engaged with the game. To acquire these rewards, a player needs to log into the game daily and complete various quests. The time-consuming nature of said quests and the possible insufficiency of the rewards might be why many players choose to use real money to buy hard in-game currency.

Arknights may be considered quite generous in this regard. The daily and weekly rewards reach 1,200 in soft in-game currency per week. Additionally, players are rewarded upon completing particular challenges, which account for a maximum of 1,800 in said currency (Arknights, 2020). The 3,000 units of currency provide for exactly 5 rolls per week (Arknights, 2020). Daily and special task missions are not the only way to acquire these resources. However, unlike in Genshin Impact and Faith/Grand Order (Genshin Impact, 2020; Fate/Grand Order (English), 2015), we can consider these missions the primary source of virtual currency in Arknights (Arknights, 2020).

### 1.4.4 Virtual currency

Virtual currency, also known as in-game currency, serves the same purpose as any other currency. It can be exchanged for an item of in-game value. The economic system of gacha games is isolated from the real economy in the sense that neither the currency nor the reward can be transferred back to money, we can see this in figure 1.2. Selling accounts is considered illegal as it would connect the gacha system back to the real economy and classify it as a game of chance, implying more regulation on the gacha game.

Virtual currency is an essential part of the gacha model and provides a way to monetize the game. Arknights, just like Genshin Impact, has two basic currencies related to its gacha system (Arknights, 2020; Genshin Impact, 2020). Soft currency is earned by playing, while hard currency can be obtained through monetary purchases. Hard currency often must be transferred to the currency that can be used to enter the gacha system. Arknights and Genshin Impact (Arknights, 2020; Genshin Impact, 2020) use this approach. In Genshin Impact, there are multiple rolling currencies that depend on a banner type. Once transferred, the rolling soft currency can not be changed back into hard currency (Arknights, 2020; Genshin

*Impact*, 2020) or a different kind of soft rolling currency. (*Genshin Impact*, 2020). The value of duplicate rewards can be connected to the virtual currency. The reward can provide either additional currency (*Genshin Impact*, 2020) or shop currency (*Arknights*, 2020).

The system of in-game currencies is usually more complicated. There are other types of currency serving in-game purposes not directly related to paid or soft currency and gacha system. This work only considers the types directly related to the gacha system.

#### 1.4.5 Limited time events

An essential part of the gacha model are limited time events. In some video games like *Faith/Grand Order* or *Genshin Impact*, limited time events rewards may be considered the main source of virtual currency (*Fate/Grand Order (English)*, 2015; *Genshin Impact*, 2020). In *Arknights*, these limited-time events reward players with various items and sometimes even a character that cannot be acquired using the gacha system (*Arknights*, 2020). In terms of in-game currency, these events provide rewards that can be used to buy rolls and use the gacha system. However, they are not the primary source of virtual currency as the game offers particular game levels that reward players with said currency (*Arknights*, 2020).

There can be, and often are, differences in the usual gacha mechanics and mechanics for the banners connected to the limited time events. Players are encouraged to spend more virtual currency on rolling, as they can get the reward only for a limited time. Some games include an additional pity system for these events to reduce the randomness and increase the fairness of the gacha system. Research done in this area by other people is focused more on game psychology, or economic models may have more information on the differences in perception of these events. For most, the mechanisms stay the same. The simulation created in this work discusses the standard banners.

#### 1.4.6 Virtual items, characters and other rewards

Players participate in gacha in order to gain something of in-game value. This reward can be an item, a character, or a similar object of in-game value. The reward can have many forms, but there are some characteristics it typically fulfills. As mentioned in the paper by Koeder and Tanaka (Koeder and Tanaka, 2017), and based on the author's observation, we can define the following characteristics:

- **has a role in a game** – in-game value that can be decorative, functional, or social,
- **duplicates have value** - rewards do not lose value once acquired and can be used to improve the characteristics of the original reward; upon reaching the maximum, it can still be converted to something of in-game value,

- **varied rarity** – every reward belongs in a one rarity category, the number of categories usually not less than three and no more than six, higher rarity has lower chances of obtaining and visa versa,
- **non-monetary** – virtual items acquired in the game are only available in the game and cannot be traded for real money,
- **limited** – some or all of the virtual items can be limited, meaning players can only acquire them during a limited time,
- **part of an event** – some virtual items are connected to an in-game event,
- **collectible**.

### 1.4.7 Banners

Banners change probabilities on specific rewards and can also change the pool of rewards. Gacha games can provide multiple banners that often change (rotate) over time, but the presence of a permanent banner is common. There are also limited and special banners.

An excellent example of this can be seen in Genshin Impact. Three different kinds of banners exist in Genshin Impact (*Genshin Impact*, 2020). A standard banner is available at all times. Limited character banners include characters with increased chances of obtaining (rate up), similar to the standard headhunting banners in Arknights. Limited character banners in Genshin Impact also include a special pity system that guarantees the rate up character to be the second one in case the first one is a different highest rarity reward (*Genshin Impact*, 2020). On the other hand, standard headhunting banners in Arknights do not include any pity system targetting specific operator (*Arknights*, 2020).

### 1.4.8 Stamina

As players progress by playing, there is a need to limit the amount of rewards they can obtain. This limitation can have various names in different gacha systems, but its core purpose remains.

Sanity can be seen as a bar of energy that is being restored over time. It can serve various purposes and is often rechargeable through in-game currency and by an increase in the player's level (this can also increase the top limit of this bar). Players usually spend their sanity either by playing or directly collecting the rewards.

## 2. Probability theory in gacha system

In this chapter, the author will build upon the current knowledge by placing this bachelor's thesis into the theoretical framework. By putting this work into perspective, the author hopes to not only gain a decent understanding of statistics and probability theory necessary for the practical part but also aims to introduce important terms related to this work.

The theory of probability is highly relevant to this work. It provides a tool to model and analyze several phenomena. Being able to model complex systems may even be the main reason to study probability and random process (Miller and Childers, 2012). Therefore in the section 2.1 probability is defined and its relevance to this paper is explained. Section 2.2 introduces the term random experiment. Next, section 2.3 introduces stochastic simulation and related terms that include the methodology. Last but not least, section 2.4 looks into the bases of probability in gacha systems, especially in the game Arknights.

This chapter explains only the fundamental of probability theory and highlights the application to the simulation created in the next chapter. For more information on these topics author refers to cited work and other literature. This area is well researched and going into details is not the purpose of this work.

### 2.1 Probability

Probability is an essential part of the gacha mechanism and the simulation author aims to create. Migon, Gamerman and Louzada even claim in their book *Statistical Inference: An Integrated Approach, Second Edition* that the concept of probability was first formulated based on the ideas closely related to the game of chance, such as cards, dice or coins (Migon et al., 2014).

There are multiple perspectives and possible interpretations of probability. For this work, a sufficient definition is that probability represents a ratio between the number of specific outcomes and the total number of possible results of a random experiment (Migon et al., 2014). In the simulation presented in chapter 3, a representative sample of outcomes is created based on probabilities.

### 2.2 Random experiment

According to Blitzstein and Hwang, a random experiment is any process or procedure that produces a set of possible outcomes that are uncertain or unpredictable (Blitzstein and Hwang, 2015). In a book *Probability Theory and Statistical Inference: Econometric Mod-*



eling with *Observational Data* by Aris, a random experiment is specified by the following conditions (Aris, 1999):

- all possible distinct outcomes are known before the experiment,
- the outcome of any particular trial is not foreknown,
- regularity of occurrences associated with outcomes can be observed,
- repeatable under identical conditions.

Based on these characteristics, both the Gacha system and the simulation this work aims to create can be considered a random experiment. In this case, it is a type of random experiment where a computer is used to generate a set of representative possible outcomes.

## 2.3 Stochastic simulation

Simulations are appropriate means to approach both systems with all their components known and systems where some or all of their elements are subject to random fluctuations. In these cases, we speak of stochastic simulations (Gamerman and Lopes, 2006). Simply put, stochastic simulations are computational simulations that incorporate randomness or uncertainty in the input parameters or the model itself. The objective of the simulation is to reproduce random variables in an environment under our control. These simulations are possible thanks to the existence of random and pseudo-random number generators, which are the first step in creating the stochastic simulation. (Gamerman and Lopes, 2006).

### 2.3.1 Stochastic process

The definitions of the stochastic process vary, but they generally agree that it's an indexed collection of random variables. For this work, we will use a definition by Aris from the *Probability Theory and Statistical Inference* book. A stochastic process is an indexed collection of random variables defined on the same probability space (Aris, 1999).

#### Random process

The term random process is often used as a synonym for the stochastic process. The reason behind this appears to be the similarities in their definitions. The main difference seems to be that instead of an index as it is in the stochastic process, there is a function of the elements of a sample space, as well as another independent variable in the random process. This independent variable is often time, but in the case of performed simulation, this variable will be the roll number. In a random experiment, the random process maps each possible outcome to the function of an independent variable, as specified by some rule (Miller and Childers, 2012). Because this independent variable can be an index, we can consider both these terms,

random and stochastic process, equal. Both these terms provide accurate description of the simulation created in chapter 3.

### **Probability space**

In the context of stochastic processes, a probability space is a mathematical framework that defines the possible outcomes and probabilities associated with a random experiment. It consists of three components:

- a sample space - the set of all possible outcomes of the experiment,
- a set of events - subsets of the sample space that correspond to specific outcomes or combinations of outcomes,
- a probability measure - assigns a probability to each event in the set of events.

In the simulation this work aims to create, we consider sample space as the set of all the possible rarities and operators that the simulation selects out of. The set of events is a set of specific combinations of outcomes of the simulations. The set of events needs to be collected, saved, and analyzed by the end of each simulation scenario. Every roll and its outcome in the simulation is an event. This event includes roll number (independent variable), rarity, and specific operator. The probability measure has initial values, but as the simulations progress, it can change according to the rules, the event, and the current state of the process.

### **2.3.2 Markov process**

Markov process is a class of random processes that possess a certain characteristic that could be described as memorylessness (Miller and Childers, 2012). These processes have been extensively studied because they appear in numerous applications, including queuing systems, computer communications networks, biological systems, and a variety of other applications (Miller and Childers, 2012). It is a process that satisfies the Markov property. Discrete values of the Markov process in discrete time are known as Markov Chain (Miller and Childers, 2012). Although the gacha mechanism and its simulation are not working with time, they satisfy the Markov property. There are different types of Markov process, one commonly referred to as a Markov decision process. This type of Markov process uses an independent variable that represents the action taken by a decision maker instead of time. In the actual gacha system, the decision maker is the player. In our simulation, the scenario we follow determines the number of rolls.

### **Markov property**

The Markov property simplifies the analysis of stochastic processes. The Markov property is a mathematical property that describes a stochastic process where the future state of the

process only depends on the current state, and the future is independent of the past (Miller and Childers, 2012). In the simulation created by the author of this thesis, all the variables controlling the next event are stored where the process currently is. These variables include a roll that keeps track of the number of rolls, assigned pity as a counter, pity changing a set of rules related to roll number, and a pity that stores the assigned probabilities in its value. We could also consider a cost that serves only an informative character, the number of rolls that represents the initial value selected for each simulation scenario and does not change, and a stop that stops the process once certain conditions are met. A set of rules is created to generate new event based on these values. We can refer to the specific values of these variables as the current state of the system. The current state is one instance of state space.

### **State space**

State space determines the range of values that a variable being sampled can take. In contrast to the sample state, instead of referring to all the possible outcomes, it represents all possible states that the system can be in at any given time. The applications of state space in the simulation were discussed in 2.3.2. The simulation also depends on the initial conditions closely explained in section 3.3 of the next chapter.

### **2.3.3 Markov Chain Monte Carlo method**

In a traditional Monte Carlo simulation, random samples are generated independently of each other from a known probability distribution. The results are then used to estimate some statistical quantity of interest.

In contrast, Markov Chain Monte Carlo (MCMC) is a class of algorithms that uses the Markov Chain property to generate correlated random samples from a target probability distribution. The algorithm starts at a random or defined point in the sample space and moves sequentially to other points, generating a chain of dependent samples. The resulting chain is designed so that the probability of transitioning from one state to another depends only on the current state and not on any previous states. After a sufficient number of samples have been generated, the resulting distribution is used to estimate the desired statistical quantity.

To apply MCMC in this work, it is necessary to specify the state space. It is necessary to specify the initial probabilities and a set of rules that change the probabilities based on the current state. One chain of values is one run of the simulation, meaning multiple chains are generated depending on the number of times the simulation is executed. Initial probabilities are set based on the exact values listed in the game.

## 2.4 Probability in gacha games

Probability bases are different depending on the specific implementation. There is an unwritten design rule all the Gacha games follow and have in common. The following paragraph describes this design pattern.

The prize pool is divided into a few categories. The number differs, but it is usually not less than 3 and no more than 6. Each category consists of specific rewards, or said the other way around, every reward belongs to exactly one category. Every category has its given probability.

We can only proceed to the practical part because probabilities have to be disclosed for any application to be available from widely used and trusted sources, namely, GooglePlay (Google LLC, 2023) and AppStore (Apple Inc, 2023). Both of them contain the same message. Therefore, we can look at the guidelines of AppStore. *Apps offering “loot boxes” or other mechanisms that provide randomized virtual items for purchase must disclose the odds of receiving each type of item to customers prior to purchase* (Apple Inc, 2023). Besides this, there are also local regulations depending on the country, or self-regulation of the company. Games implementing the gacha mechanism usually disclose these odds on every banner in the section details or similar section for the reason that either the chances or possible rewards can vary.

### 2.4.1 Arknights gacha system

Because Arknights is the game whose gacha mechanism is the subject of the simulation, it is required to consistently define its rules and regularities. The gacha system in Arknights is called *Headhunting* (Arknights, 2020). There are different ways to use this mechanism. It is up to the player preferences what banner they partake in and how many resources they spend. There is also a process called *Recruitment* that is not a usual part of gacha games and can be seen as a different form of random reward mechanism with its own set of rules and probabilities. This work focuses on the *Headhunting* process.

As explained in part 1.4.7, banners can affect the probability rates of specific operators. There are rules that are valid for a group of banners. In Arknights, these rules are referred to as *Standard Headhunting*, and a player can find these rules after displaying details on any standard banner, this can be seen in figure 2.1 (Arknights, 2020). As discussed in 1.4.7, special and limited banners can have some or all of these rules customized. For the simulation model, choosing a specific banner whose exact system is to be observed, simulated, and analyzed is necessary. The author selected a specific standard banner available at the time on the global server. All the following figures, paragraphs, and the simulation, contain values related to this banner.



Figure 2.1: Standard Headhunting rules (screenshot from the game Arknights (Arknights, 2020) by author)

The rules of *Standard Headhunting* are in figure 2.1. These rules apply to most banners, including the one simulation in chapter 3 is based on. Therefore it is necessary to understand them. Looking at the figure 2.1, we see the following text: *If you have not obtained any 6-star Operator after 50 rolls, the odds of getting 6-star Operator will be increased from 2% to 4% in the next roll ...* (Arknights, 2020). With every roll where no operator of rarity 6 is obtained, the odds keep rising by 2% until it reaches 100% chance, meaning the player will obtain an operator of the highest rarity for sure by the turn 100. Once the player gets any operator of rarity 6, the odds are reduced back to 2%.

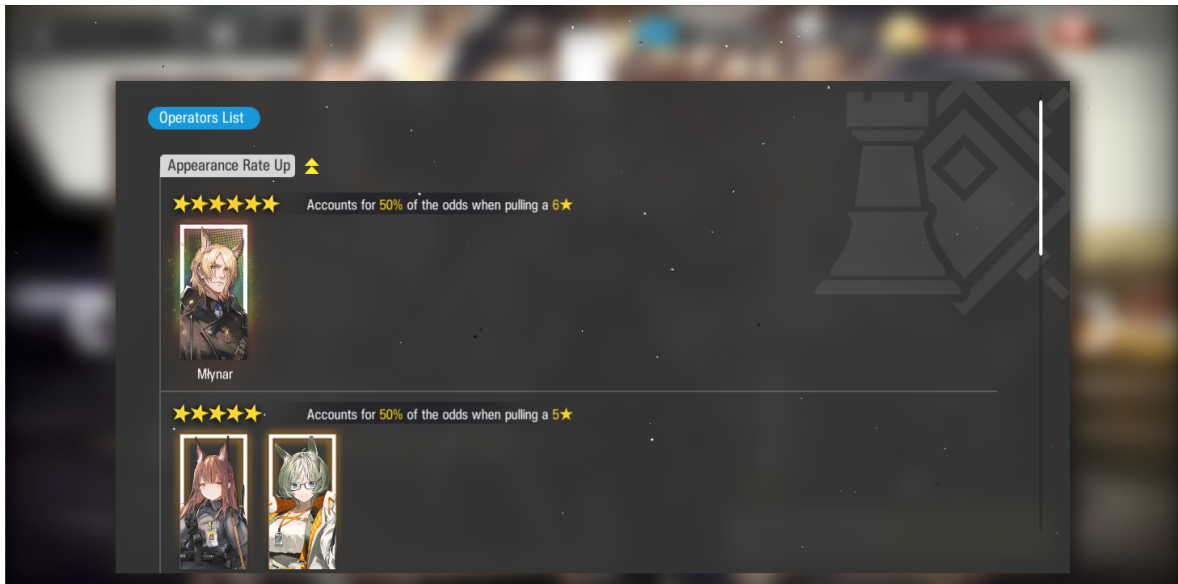


Figure 2.2: Increased chances for specific operators (screenshot from the game Arknights (Arknights, 2020) by author)

Unlike in Faith/Grand Order (*Fate/Grand Order (English)*, 2015), the pity system on standard banners in Arknights does not target one specific operator but rather the whole rarity (*Arknights*, 2020). Specific operators can be excluded from the banner, or on the other hand, their chances can be increased. Increased chances for specific operators can be seen in figure 2.2. The operators Mlynar, Franka, and Proviso have increased chances.

There is also a complete list of all possible rarities and operators in the details, this represents the sample space for the simulation. The listed values depend on the banner and the time of its availability. The list of possible rewards gets updated with more operators being added to the game. It is also different depending on the server. In this work, the global server and its banners are considered. To be more specific, we work with the above mentioned banner and its list of operators. We can see the list of possible rewards in figure 2.3. The complete list is also part of the source code in the appendix A.

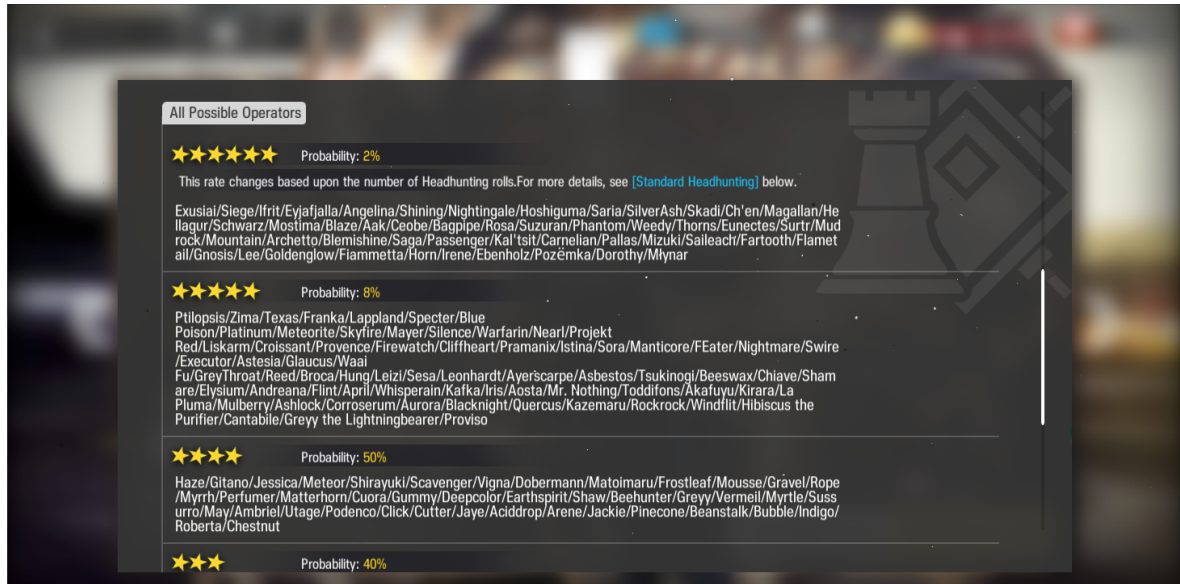


Figure 2.3: List of possible operators (screenshot from the game Arknights (*Arknights*, 2020) by author)

It is also important to mention the guaranteed rarity 5 or higher operator in the first ten rolls on every banner in Arknights (*Arknights*, 2020). A similar mechanism is in Genshin Impact (*Genshin Impact*, 2020), where every ten rolls guarantee a character of rarity 4 or higher. In Arknights, as figure 2.4 shows, this applies to every banner only once.



Figure 2.4: Guaranteed operator of high rarity (screenshot from Arknights (*Arknights*, 2020) by author)

# 3. Simulation of gacha system

This chapter built upon the probability bases, methodology, and rules of the Arknights gacha system explained in the previous chapters. The simulation model that describes the gacha mechanism shows expected development and predicts the number of resources needed for getting the selected operator of the highest rarity is created in this chapter, and its results are analyzed. The complete source code is available in the appendix A.

First, in section 3.1, tools used for the simulation are specified. Section 3.2 describes the methods used to ensure the reliability of the simulation model. Section 3.3 introduces the initial conditions, values, and initial state of the random experiment. In section 3.4, the different simulation scenarios are presented, and their results are analyzed and interpreted. Finally, in section 3.5, the results of simulations are summarized and compared.

## 3.1 Software used for simulation

The impact of the software used on simulation results can depend on the type and complexity of the simulation. It is not public knowledge what software is used by the developers of the game Arknights. Neither is it known what methods and approaches to probability were applied in the Arknights gacha system. Based on the common practice in the game development industry, we can expect them to use some of the many programming languages available. Therefore the author of this work decided to use Python as it offers appropriate tools for creating simulations and analysis of results. Still, it is important to note that there may be differences in the simulation and gacha system in Arknights.

### Software specification

The simulation was created in Python 3.10.7. As explained in 3.1, the author decided to use the pseudo-random number generator provided by the package *random*. A complete list of all used packages can be seen in the appendix A. However, besides the package *random*, they serve the mere purpose of data analysis and visualization of results.

### Random number generator

Overall, computers are not great at generating random values. Instead, when talking of random number generators, we rather speak of pseudo-random number generators (Gamerman and Lopes, 2006). Considering the character of this simulation pseudo-random number generator is a sufficient solution.

Python has multiple ways to approach randomness. One can create their own random number generator or use a package that supports this functionality. For the purpose of this work, the author identified most suiting this problem the implementation of randomness in the packages *numpy* and *random*. Both of these models have similar functionality, therefore author decided to use the package *random*, which is part of the Python standard library. The specific random experiment can be reproduced by stating the value of a seed, but there was no need to use this functionality, especially since we sometimes run hundreds of simulations in a row. There is also the option to get closer to real randomness by using the system data as input. We don't have such a high requirement for randomness, and we can expect the developers of Arknights to use a similar approach to the one we use.

## 3.2 Testing

To ensure the simulation works correctly, the author executed it multiple times with different conditions and initial values than for the analysis. The code also contains error warnings that are raised in case a function does not return valid values. Unnecessary for the main functionality of simulation are also print statements that serve as the fastest feedback about the simulation results.

In testing, the author emphasizes the need to test the functionality of the pity system. The condition of acquiring the highest rarity operator in the first 10 runs of the process was verified by observation of multiple simulation runs. Next, it was necessary to test the soft pity. Soft pity activates after fifty rolls without the highest rarity being rolled. In order to test this, it was necessary to change the probability of the highest rarity being drawn to 0%, making it temporarily unavailable. Pity then activates in roll fifty and increases the probability of the highest rarity being picked in the next roll to 100%, ensuring the highest rarity is selected right after activation of pity. This correct output can be seen in table 3.1 below. In the first three columns, there are the data we want to collect, the index starting with 1 representing the number of rolls followed by selected rarity and specific selected character. The 4th column represents the value of the pity counter. Once the pity counter reaches 50, a function that changes the set of probabilities activates. Once rarity 6 is selected, the system returns the probabilities to the original value.

Table 3.1: Test results (by author), pity test equals the value of probability for the highest rarity being selected in next roll

| Roll number | Rarity | Name           | Pity count | Pity test |
|-------------|--------|----------------|------------|-----------|
| 49          | 4      | <i>Jessica</i> | 49         | 0.00      |
| 50          | 5      | <i>Provi</i>   | 50         | 1.00      |
| 51          | 6      | <i>Mlynar</i>  | 0          | 0.00      |
| 52          | 4      | <i>Greyy</i>   | 1          | 0.00      |



For those interested in a more precise examination of the script and its functionality, the code included in the appendix A can be utilized to conduct customized tests. Particularly in the development of more complex projects, it is advisable for testing to be performed by a separate individual from the author. Given the size of this project, the aforementioned testing protocol is deemed sufficient.

### 3.3 Initial values

Every run of the simulation starts in the same state with initial probabilities and specified state space. There are initial values related to rarity in the table 3.2. Every simulation also starts in the state where the change is set to False and pity is set to 0.02. Both of these variables control different parts of the pity system. While pity represents the probability of rarity 6 being drawn, change controls the first pity condition. A number of rolls is also included and its initial value depends on the number of runs we want to perform.

Table 3.2: Initial states based on given values  
in the game Arknights (by author)

| Rarity   | Probability | Possible draws |
|----------|-------------|----------------|
| Rarity 3 | 0.40        | 16             |
| Rarity 4 | 0.50        | 40             |
| Rarity 5 | 0.08        | 70             |
| Rarity 6 | 0.02        | 50             |

The simulation consists of a given number of rolls. For each roll, we observe its number/index, the selected rarity, and a selected operator. For the first 10 rolls, it is also important to keep track of whether the rarity 5 or 6 has been drawn. There also needs to be kept track of the value of pity in simulation as it changes the initial probabilities.

### 3.4 Results

The complete Python source code is available for viewing in the appendix A. As various simulations are executed, and conditions are altered, multiple facets of the Arknights gacha system will be observed and scrutinized. The alterations in conditions will be explicated at the outset of each specific simulation scenario. Every simulation experiment is created on the bases of one specific banner that was available at the time this work was created. The operator with increased chances on this banner is Mlynar.

### 3.4.1 Simulation with the complete functionality

The following simulations proceed with the same simulation scenario but with a different number of iterations. They implement the complete functionality, including the pity system and stop condition once the operator of the highest rarity with increased chances (Mlynar) is selected.

#### Sample data

The sample data aims to present the functionalities and data processed and returned by the simulation rather than to serve as statistical measures.

The first simulation returned the desired outcome in the first roll. Despite the mere chance of 2 %, this outcome can occur. This set of values is not suitable for a description of the functionalities, nor does it provide good exemplary value as this situation is not very likely to occur. We can see this in graphs and on statistical measures discussed in the following two sections, where we run the same simulation 100 times and then 10,000 times.

Sample results of one simulation more suitable for initial description are in the table 3.3 for the first 10 rolls. The rarity distribution for the rest of the rolls can be seen in the following figures 3.2 and 3.1.

Table 3.3: Sample results of first 10 rolls of one simulation (by author)

| Roll number | Rarity | Character        |
|-------------|--------|------------------|
| 1           | 3      | <i>Popukar</i>   |
| 2           | 4      | <i>Sussurro</i>  |
| 3           | 4      | <i>May</i>       |
| 4           | 4      | <i>Deepcolor</i> |
| 5           | 4      | <i>Vermeil</i>   |
| 6           | 3      | <i>Plume</i>     |
| 7           | 4      | <i>Cutter</i>    |
| 8           | 4      | <i>Cuora</i>     |
| 9           | 3      | <i>Fang</i>      |
| 10          | 5      | <i>Proviso</i>   |

There is no reward of rarity 5 or 6 in the first nine rolls. As mentioned earlier, this simulation model implements the pity system. Therefore the 10th reward has to be rarity 5 or 6. This can be seen in the 10th row of table 3.3, where the output of the simulation is an operator of rarity 5, Proviso. The chances of selecting rarity 5 over rarity 6 represent the default ratio between them, meaning an 80% chance of rarity 5 and 20% chance of the highest rarity.

There are six operators of rarity 4 in the first 9 rolls. Based on the initial set of probabilities

and observations on a large data sample further in this thesis, there is more likely going to be only five of them and one more operator of rarity 3 or operator of rarity 5. These nuances in isolated results of individual simulations are to be expected. In order to achieve an accurate description of the system, the random experiment needs to be conducted multiple times. For this reason, a large sample of data from multiple simulations is generated and analyzed in the consequent subsections.

Because the majority of players may use the gacha system intending to obtain a specific reward. The simulation takes this into consideration and only stops once the desired outcome is achieved. This can be seen in the following figures 3.2 and 3.1 that display the rarity distribution for the complete simulation.

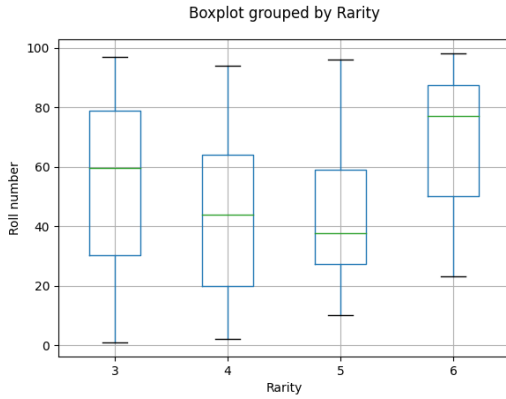


Figure 3.1: One simulation (by author)

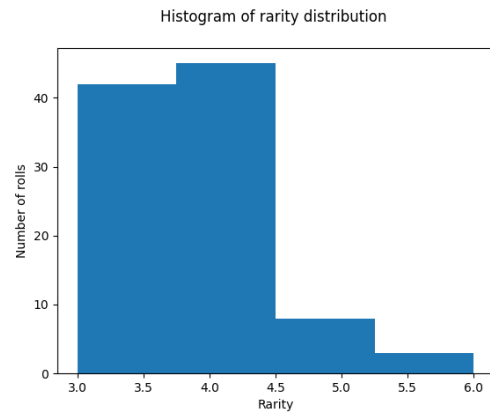


Figure 3.2: Rarity distribution (by author)

The simulation finished in 98 rolls. The highest rarity was rolled 3 times, but only the last roll returned the character named Mlynar, the target value. This stopped the simulation.

Every roll costs 600 of soft in-game currency, which is called Orundum in the game. The total cost of rolls in this simulation is 58,800 Orundum. This equals five months' worth of the standard login and task rewards (*Arknights*, 2020). This indicates that there could potentially be a problem for the player with the Arknights gacha mechanism, not in the distribution of rarity, but in the fact that the target reward on this banner always accounts for exactly 50 % of the odds when pulling a rarity 6. To test this, the random experiment will be repeated multiple times, and based on the analysis and observation, we will answer this.

Players could encounter similar problems when targeting an operator of rarity 5. In this simulation, the first time rarity 5 is rolled is in the 10th pull. There are two operators with increased chances in this rarity category, Proviso, and Franka. While in table 3.3 we can see Proviso being selected in the 10th roll, Franka was selected only two rolls before the simulation finished. Compared to the highest rarity, figure 3.1 shows that rarity 5 was rolled rather sooner with the lowest median value out of all the rarities. Figure 3.2 shows that rarity 5 was rolled only eight times, less than lower rarities, but more than twice as much compared to rarity 6.

For the reasons explained above, the author decided to study the highest rarity and occurrences connected to the character with increased chances, Mlynar. Similar rules may apply to the rarity 5 but to a lesser extent.

### 100 simulations

Results presented in this part are created based on 100 simulations with the same conditions as the previously studied example. One hundred simulations, with a maximum of 1,000 rolls for each, have been run and analyzed in this section.

In the previous subsection, we stated the need to observe the simulation behavior related to obtaining the highest rarity reward with increased chances, operator Mlynar. Because this is also the condition that concludes the simulation, in figure 3.3, there is a box plot showing the simulation results. The value of the result represents a roll number in which the operator Mlynar was rolled for every simulation. None of the simulations finished without achieving their goal. The box plot in figure 3.3 shows that most simulations end in not even 100 rolls. One of the outlier values is reaching as high as 555 rolls. Therefore the outliers should not be ignored.

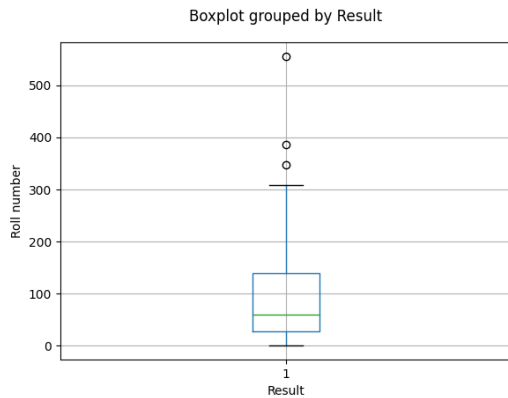


Figure 3.3: Box plot for final rolls in 1,000 simulations (by author)

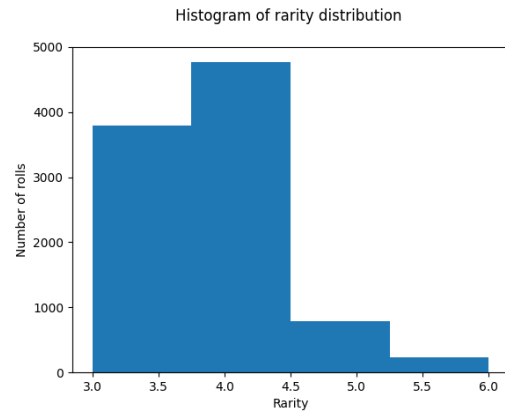


Figure 3.4: Rarity distribution in 1,000 simulations (by author)

The best-case scenario, having a 2% chance, is achieving the desired outcome in the first roll. This best-case scenario was more likely to occur than a simulation that did not return the desired outcome in over 300 rolls. The chances of not obtaining Mlynar in 6 pulls of the highest rarity in one simulation are as low as 1.5625%, and with every roll that returns the highest rarity, this keeps becoming more unlikely.

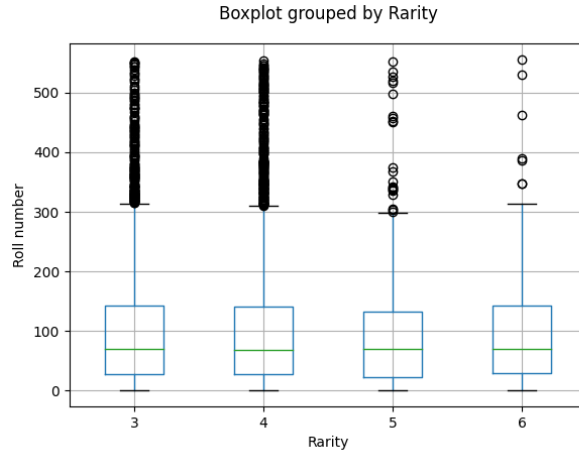


Figure 3.5: Rarity in 100 simulations (by author)

Table 3.4 and figures 3.3, 3.4, and 3.5 summarise the statistical measures of the generated sample. All the simulations finished before reaching the ceiling of 1,000 rolls. Despite this being an apparently positive result, the author needs to note that the absolute majority of players do not have this amount of in-game currency on hand, and obtaining it through monetary purchases for the outliers would be extremely expensive. Therefore for players who do not necessarily need the operator with increased chances, it could be advisable to stop after a series of unsuccessful pulls of different rarity 6 operators.

Table 3.4: Statistical characteristics of 100 simulations  
(by author)

|      | Roll number | Total cost     | Result |
|------|-------------|----------------|--------|
| mean | 95.840000   | 57,504.000000  | 1.0    |
| std  | 97.885701   | 58,731.420737  | 0.0    |
| min  | 1.000000    | 600.000000     | 1.0    |
| 25%  | 26.750000   | 16,050.000000  | 1.0    |
| 50%  | 60.000000   | 36,000.000000  | 1.0    |
| 75%  | 139.500000  | 83,700.000000  | 1.0    |
| max  | 555.000000  | 333,000.000000 | 1.0    |

Only 100 chains of values created by the simulation may not describe all the variances of results that can occur. To ensure the accuracy of results, we follow this experiment in the next part, where we run a total of 10,000 simulations.

### 10,000 simulations

Without the further need to introduce the conditions of this simulation, this part presents the results of 10,000 simulations.

The histogram 3.7 of rarity distribution in 10,000 simulations and the histogram for rarity distribution in figure 3.4, from the previous subsection, have a very similar shape that also fits the initial probability space. This observation confirms that the equal number of occurrences of rarities 3 and 4 in the sample simulation was one of the many possible nuances between individual runs of simulation caused by the nature of random experiments.

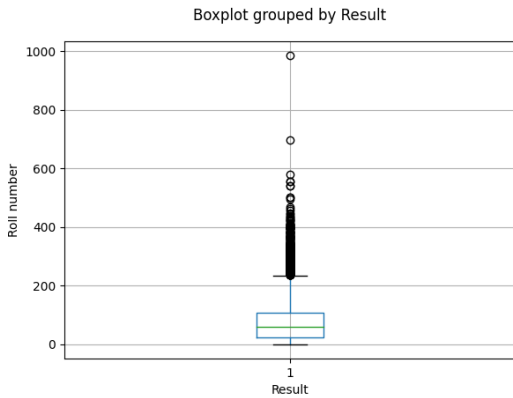


Figure 3.6: 10,000 results of simulations (by author)

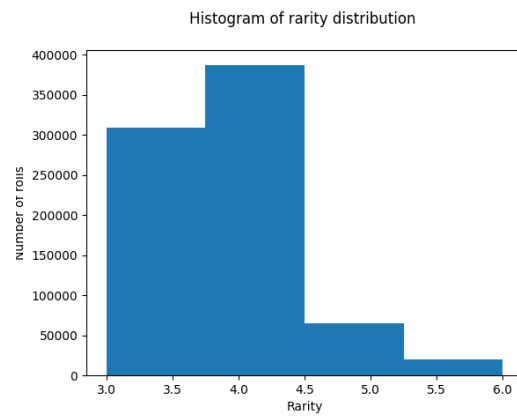


Figure 3.7: Rarity distribution in 10,000 simulations (by author)

The maximal value representing the end of a simulation is 985. The maximum, alongside the minimum, shows that the range of results is quite extensive, especially after considering the cost of each roll. Table 3.5 displays this together with other statistical characteristics. The maximal value is very close to the ceiling of the simulation. To see whether this is a rare case, we can look at the table of statistical measurements or see the results visually represented in the box plot 3.6.

Table 3.5: Statistical characteristics of 10,000 simulations (by author)

|      | Roll number | Total cost     | Result |
|------|-------------|----------------|--------|
| mean | 78.115100   | 46,869.060000  | 1.0    |
| std  | 72.252742   | 43,351.645081  | 0.0    |
| min  | 1.000000    | 600.000000     | 1.0    |
| 25%  | 25.000000   | 15,000.000000  | 1.0    |
| 50%  | 59.000000   | 35,400.000000  | 1.0    |
| 75%  | 109.000000  | 65,400.000000  | 1.0    |
| max  | 985.000000  | 591,000.000000 | 1.0    |

In table 3.5, we can see the average value of the simulation result as well as the median. While the median and overall percentile distribution did not change much compared to table 3.4, the average value makes a difference in over 15 rolls. This inconsistency in the average value and, on the other hand, stable percentile distribution indicates that the median may have a higher informative value for the player, as it shows in which roll 50 % of players achieve the desired outcome.

Despite the relatively low modus, players should also be aware of the outliers and the possibility of being one of the few players that achieve the desired result later as the values are spread around the mean. This is also suggested by the value of standard deviation (std) of the roll number, which is lower than the one in the data from fewer simulations in table 3.4. Still, this high value suggests that the end rolls of the simulation are relatively spread out around the mean. These results all together achieved sufficient accuracy and informative value. Therefore for further analysis, 10,000 simulations will be conducted.

### **3.4.2 Simulation of the first 10 rolls on a banner**

On account of the guaranteed operator of rarity 5 or higher in the first 10 rolls on any banner in Arknights, this part of Arknights gacha system is interesting to observe using simulations. We can approach this easily by running the simulations only for the first 10 rolls.

This system targets rarity 5 and 6. In this work, we also call the group consisting of rarity 5 and 6 a high rarity. The first simulation and its results will be described, much like the simulation in the section 3.4.1. In fact, the results of these two sample simulations can be compared for the first 10 rolls.

After the sample simulation, there will be two scenarios in this section. First, we will take a look at simulations consisting of precisely 10 rolls. Later, the situation when the simulation stops after it returns an operator of the high rarity will be observed. Both of these will apply the complete functionality specified in section 3.4.1. These results can be compared in section 3.4.3 with results of 10 rolls in simulation without the pity system.

#### **Sample data**

There are 4 operators of rarity 3 and five operators of rarity 4 in the first nine rolls. Because there is no higher rarity, the last roll could be only the high rarity. In this simulation, it is a rarity 6, operator Młynar. We can see the rarity distribution in the following figures 3.9 and 3.8. Later, it will be possible to compare this with the outcomes of different simulation scenarios based on 10,000 simulations.

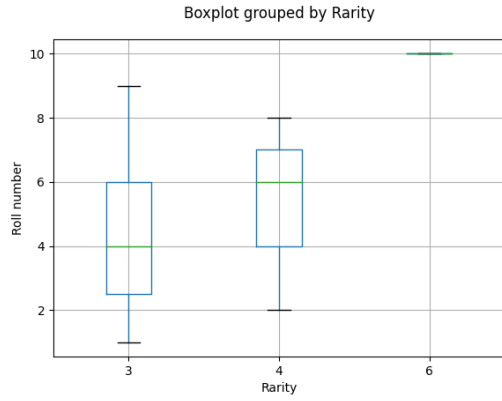


Figure 3.8: Rarity in 10 rolls, one simulation (by author)

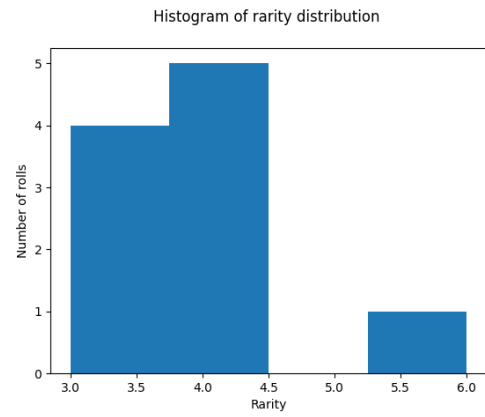


Figure 3.9: Rarity distribution in 10 rolls, one simulation (by author)

Based on the simulation as well as the set of probabilities and observation, the pity mechanism for the first 10 rolls does activate in less than half of the observed cases. High rarity tends to be selected before its activation, but it is a good way to ensure 10% chance in 10 separate cases turns into 100% chance on the 10th run through the random process as it increases the overall chances of obtaining high rarity reward in the first 10 rolls.

### Simulations of the first 10 rolls

Simulations which results are presented in this part implement the complete functionality, although the pity system that activates after fifty rolls without rarity 6 selected has no means to activate. The following data consists of 10,000 simulations, each with 10 values that account for 10 rolls.

The figures 3.10 and 3.11 show rarity distribution. Rarity 6 appeared 2,677 times, and in 1,342 cases, it is the operator with increased chances. This ratio perfectly describes the 50% probability of the rate up operator when selecting the rarity 6.

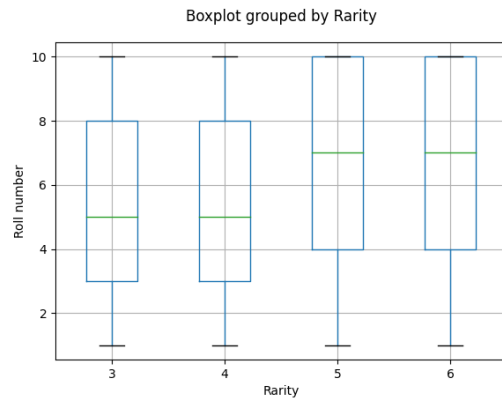


Figure 3.10: Rarity in 10 rolls (by author)

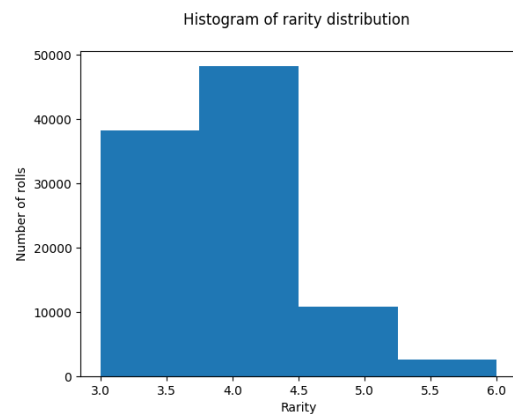


Figure 3.11: Rarity distribution (by author)



If we wanted to calculate the probability of getting rarity 6 from this sample, it would be 2.677 % for each roll. This probability is higher than the initial 2% probability of rolling the highest rarity. This increased probability indicates that the pity system influences the outcome as it increases the overall probability of getting the highest rarity operator in the first 10 rolls to about 1.34 %.

Rolling until this pity system activates may be a good practice for players to increase the chances of obtaining a high and even the highest rarity reward. We can support this by the median value for obtaining high rarity rewards, which is seven, as shown in the figure 3.10. In a system without any pity effect, the median value would most likely be around five. To simulate the system's behavior when a player stops rolling once he obtains a high rarity reward, we stop the simulation after this pity system is disabled. The simulations will finish after the pity system activates during the 10th roll or after a high rarity reward was in any previous roll. We will follow this example next.

### Simulations that stop after high rarity is obtained

In this simulation scenario, we simulate the behavior of a player that only rolls until he gets the high rarity reward. In 3,836 cases out of 10,000, the pity was activated in the 10th turn. Based on the simulation as well as the set of probabilities and observation - the pity mechanism for the first 10 rolls does activate in less than half of the observed cases - high rarity tends to be selected before its activation. This said, it is still a good way to ensure a 10% chance in 10 separate rolls turns into a 100% chance by the 10th run of the random process, increasing the overall chances of obtaining a high rarity reward in the first 10 rolls.

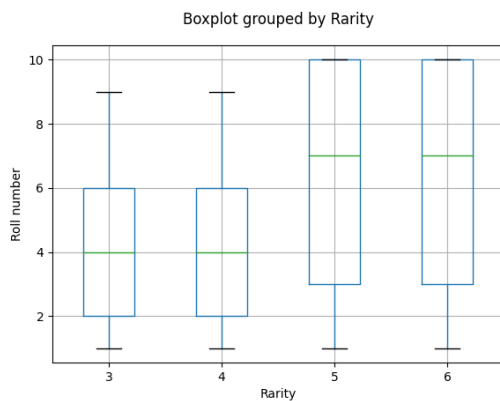


Figure 3.12: Rarity in 10 rolls, based on results of 64,939 rolls (by author)

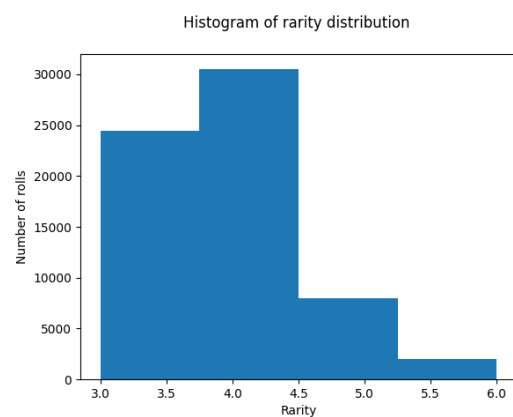


Figure 3.13: Rarity distribution for 10,000 simulations - 64,939 rolls (by author)

The figures 3.12 and 3.13 show statistical measures as well as rarity distribution. Rarity 6 was selected 2,021 times out of 64,939 rolls, indicating a probability of 3.112 %. This shows an increase compared to the base probability over 1.5 % as well as an increase over the probability of the highest reward in the simulation scenario in section 3.4.2.

From these results, it is evident that for players who necessarily do not target the specific operators on a banner and instead aim for any high rarity reward, using the pity of guaranteed high rarity operator in the first 10 rolls increases the overall chances of obtaining rarity 6 operators. This is especially evident in this simulation experiment, as we stop once a high rarity reward is obtained. Players who use this mechanism to their benefit would rather roll on many different banners instead of spending all their resources on one.

### 3.4.3 Simulations without the pity system

In a gacha system without pity, every roll is independent of any previous one. The standard Monte Carlo method can be used to simulate this simpler system. In this part, 10,000 simulations are generated, and their results are analyzed.

Because applying this to the first 10 rolls does not only disable the first pity condition but the second pity condition as well, the author does not see the need to run the simulation with a higher number of rolls and no pity condition. Unlike in the previous sections, values generated by the gacha mechanism without pity do not create a chain. Instead, they are fully independent of each other. Therefore, it does not matter how many simulations of how many rolls we run. Instead, we observe the total number of rolls. The rules were taken from the simulations of the first 10 rolls so we can have the same scale for comparison of results.

The same as explained in the chapter focused on the first 10 rolls with pity. Therefore there is no need to describe the system. After the pity system is disabled, we can proceed with this random experiment and compare the results.

#### **Simulation without pity system - no stop condition**

On the following histogram 3.15, we see the distribution of rarity in simulation consisting of 100,000 rolls without any pity condition. The results of each roll are independent of each other. Regardless, the author ran 10,000 simulations by 10 rolls for each. These conditions are set to better compare results with the simulation results in section 3.4.2 that implements the pity.

From the total of 100,000 rolls, over 10,000 are high rarity rewards. Exactly 2,033 of these cases were rarity 6, and half of these (1,017) was the rate up operator with a 50% chance. Operators of rarity 5 were rolled in 8,107 cases. The simulation without pity system most accurately represents the initial probability space not altered by pity.

In figure 3.14, we can see that the distribution of individual rarities in 10 rolls is relatively equal in every roll. We can compare these results with the results of simulations for the first 10 rolls in section 3.4.2, especially figure 3.10 that is based on the same simulation except with the pity system included. Not only do rare rarities appear less in simulations without pity, but they also do not lean toward the 10th roll.

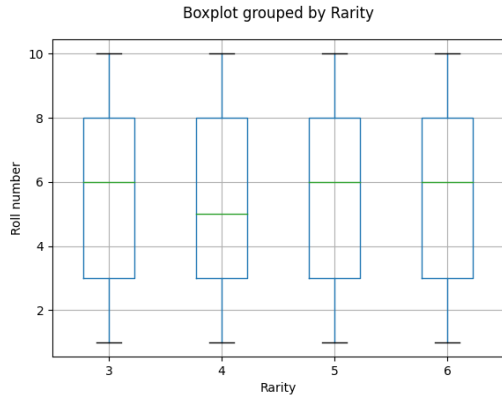


Figure 3.14: Rarity in 10 rolls without pity, based on results of 100,000 rolls (by author)

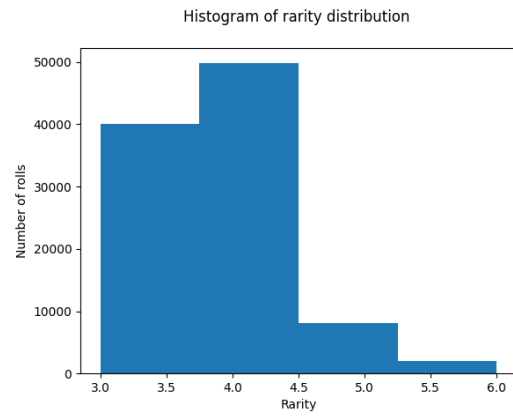


Figure 3.15: Rarity distribution in 100,000 rolls without pity (by author)

### Simulation without pity system that stops after high rarity is obtained

Rarity 6 was rolled 1,320 times out of 64,762 rolls. The number of total rolls is almost the same as those in the simulation scenario, including the pity condition. However, the highest rarity is rolled less in this scenario. Based on these results, the probability of rarity 6 being selected is 2.042 %. This is very close to the initial probability for this rarity. We can see these differences in the following figures 3.16 and 3.17 as well.

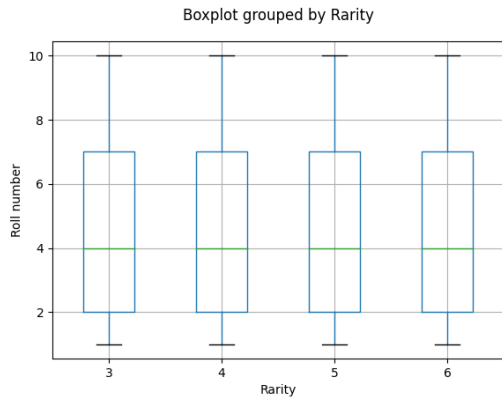


Figure 3.16: Rarity in 10 rolls without pity, based on results of 64,762 rolls (by author)

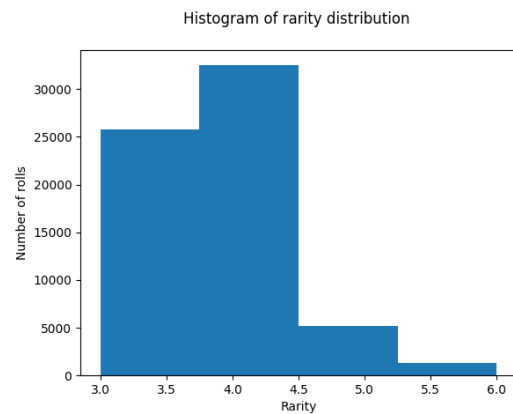


Figure 3.17: Rarity distribution for 10,000 simulations - 64,762 rolls (by author)

## 3.5 Summary of results

There is a high variance in possible scenarios that could be designed, tested, and analyzed. The source code for the simulation is available in the appendix A and can be customized and used to conduct unique simulations with contrasting sets of results. Values used in the simulation can be changed to describe separate banners or gacha mechanisms entirely.

The section 3.4.1 implementing the complete functionality shows that the number of necessary resources cannot be predicted with certainty. We can see this in table 3.6. There are noticeable differences in values depending on the number of simulations, and the high values of standard deviation show how specific simulations are distinctive despite following the same set of rules and probability space with the same initial probability function. The desired outcome, highest rarity operator Mlynar, is in half of the cases rolled by the roll number 60. This accounts for 36,000 in-game currency, which is equal to 3 months of the usual in-game login and task rewards or equal to approximately 150\$ worth of premium currency. Because of the high spread of values around the mean, as the standard deviation and outliers suggest, the individual simulations return a wide variety of results.

Table 3.6: Comparison of results in simulation with complete functionality (by author)

|      | Roll number (10,000 simulations) | Roll number (100 simulations) |
|------|----------------------------------|-------------------------------|
| mean | 78.115100                        | 95.840000                     |
| std  | 72.252742                        | 97.885701                     |
| 25%  | 25.000000                        | 26.750000                     |
| 50%  | 59.000000                        | 60.000000                     |
| 75%  | 109.000000                       | 139.500000                    |

The wide range of possible outcomes set the value of the mean quite high above the modus. The average necessary resources required for the highest rarity reward with increased chances are therefore higher. About 48,000 of Orundum, which equals to 4 months' worth of login and task rewards or approximately 200\$ worth of premium currency, is spent on average. The average values were obtained from simulations based on the situation when a player has enough resources to continue to engage with the gacha mechanism even after a series of unsuccessful rolls. Considering the limited resources of actual players, some simulations would not achieve the desired outcome, unlike the simulations presented in this work.

Table 3.7: Comparison of 10 rolls with and without pity system (by author)

|                    | With pity system |         | Without pity system |         |
|--------------------|------------------|---------|---------------------|---------|
|                    | No stop          | Stop    | No stop             | Stop    |
| Total rolls        | 100,000          | 64,939  | 100,000             | 64,762  |
| High rarity rolls  | 13,483           | 10,000  | 10,140              | 6,500   |
| High rarity chance | 13.183%          | 15.399% | 10.140%             | 10.036% |
| Rarity 6 rolls     | 2,677            | 2,021   | 2,033               | 1,320   |
| Rarity 6 chance    | 2.677%           | 3.112%  | 2.033%              | 2.042%  |

*Note:* Stop refers to simulations that stop after the pull of high rarity reward.

Based on the results of the 10,000 simulations for the first 10 rolls, presented in table 3.7, the impact of the pity system on rarity is considerable.

# Conclusion

In the first chapter, we introduced the topic of gacha system and compared three different implementations of the gacha model. The simulation of the Arknights gacha system was developed on the foundation of theoretical knowledge that was presented in the second chapter dedicated to Probability theory and its application to any but especially Arknights gacha system and its simulation.

This thesis fulfilled the given goals. The simulation of the Arknights gacha system was designed, tested, and executed multiple times in multiple defined scenarios. Results show that the average number of necessary resources for the player to acquire a reward of the highest rarity with the increased chances is about 48,000 in-game currency, this equals approximately four months of saving up the soft currency, or hard currency equal to 200\$. While the average number of rolls is this high, the mode is only about 36,000. The results indicate that in most cases, the desired outcome is achieved rather sooner, and the average number reaches such a high value, mostly for unfortunate cases. Results of the experiment observing the pity that guarantees a high rarity operator in the first ten rolls indicate that part of the reason for the mode to be lower than average may be the fact that chances for the highest rarity reward are in the first 10 rolls over 1.5 % increased compared to the initial probability.

This thesis has the following limitations. The simulation was designed and applied for a mere one existing gacha system in the game Arknights and focused mainly on one banner. Analyzing other gacha systems with simulations and comparing them would be beneficial. This work also does not include an accurate mathematical description of the performed random simulation. Neither contains sample data from the actual gacha system in the game Arknights. The description of patterns, elements, and functionalities in the gacha model was based on observations and comparisons of mainly three different gacha games. Neither the author nor this work takes a stand on whether the gacha mechanism can be considered gambling, nor does it state any judgments. The simulation results are informative and do not present a universal strategy to utilize the gacha system. The results represent multiple approaches and outcomes depending on a set of conditions.

Despite these limitations, the simulation can provide more context and a different perspective on gacha and similar random reward mechanisms. Additionally, a new gacha system could be designed based on this work. This thesis emphasizes the importance of pity in such a system. The pity system has proven to have a noticeable effect on the results. After considering the economic and psychological aspects of this phenomenon, this thesis could be useful material for developers who want to design a gacha system that would not only provide high returns for the company but also increase the satisfaction of the players with a game and especially its gacha mechanism.

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# **Appendices**



# A. Source code of simulations in Python

```
1 import csv
2 import random
3 import matplotlib.pyplot as plt
4 import pandas as pd
5
6 # Sample space
7 rarity_three = ['Fang', 'Vanilla', 'Plume', 'Melantha', '
    Cardigan', 'Beagle', 'Kroos', 'Lava', 'Hibiscus', 'Ansel', '
    Steward', 'Orchid', 'Catapult', 'Midnight', 'Spot', 'Popukar'
    ]
8 rarity_four = ['Haze', 'Jessica', 'Meteor', 'Shirayuki', '
    Scavenger', 'Vigna', 'Doberman', 'Matoimaru', 'Frostleaf', '
    Mousse', 'Gravel', 'Rope', 'Myrrh', 'Perfumer', 'Matterhorn',
    'Cuora', 'Gummy', 'Deepcolor', 'Earthspirit', 'Shaw', '
    Beehunter', 'Greyy', 'Vermeil', 'Myrtle', 'Sussurro', 'May',
    'Ambriel', 'Utage', 'Podenco', 'Click', 'Cutter', 'Jaye', '
    Aciddrop', 'Arene', 'Jackie', 'Pinecone', 'Beanstalk', '
    Indigo', 'Roberta', 'Chestnut']
9 rarity_five = ['Ptilopsis', 'Zima', 'Texas', # 'Franka',
10 'Lapland', 'Specter', 'Blue Poison', 'Platinum', 'Meteorite',
    'Skyfire', 'Mayer', 'Silence', 'Warfarin', 'Nearl', 'Projekt
    Red', 'Liskarm', 'Croissant', 'Provence', 'Firewatch', '
    Cliffheart', 'Pramanix', 'Istina', 'Sora', 'Manticore', '
    FEater', 'Nightmare', 'Swire', 'Executor', 'Asthesia', '
    Glaucus', 'Waai Fu', 'GreyThroat', 'Reed', 'Broca', 'Hung', '
    Leizi', 'Sesa', 'Leonhardt', 'Ayerscape', 'Asbestos', '
    Tsukinogi', 'Beeswax', 'Chiave', 'Shamare', 'Elysium', '
    Andrena', 'Flint', 'April', 'Whisperian', 'Kafka', 'Iris', '
    Aosta', 'Mr. Nothing', 'Toddifons', 'Akafuyu', 'Kirara', 'La
    Pluma', 'Mulberry', 'Ashlock', 'Corroserum', 'Aurora', '
    Blacknight', 'Quercus', 'Kazemaru', 'Rockrock', 'Windflit', '
    Hibiscus the Purifier', 'Cantabile', 'Greyy the
    Lightningbearer', # 'Proviso'
11     ]
12 rarity_six = ['Exusiai', 'Siege', 'Ifrit', 'Eyjafjalla', '
    Angelina', 'Shining', 'Nightingale', 'Hoshiguma', 'Saria', '
    SilverAsh', 'Skadi', 'Ch\`en', 'Magallan', 'Hellagur', '
    Schwarz', 'Mostima', 'Blaze', 'Aak', 'Ceobe', 'Bagpipe', '
    ]
```

```
Rosa', 'Suzuran', 'Phantom', 'Weedy', 'Thorns', 'Eunectes', '
Surtr', 'Mudrock', 'Mountain', 'Archetto', 'Blemishine', '
Saga', 'Passanger', 'Kal\'tsit', 'Carnelian', 'Pallas', '
Mizuki', 'Saileach', 'Fartooth', 'Flametail', 'Gnosis', 'Lee'
, 'Goldenglow', 'Fiamemtta', 'Horn', 'Irene', 'Ebenholz', '
Pozemka', 'Dorothy', # 'Mlynar'
13     ]
14
15
16 # Value that determines probabilities
17 pity = 0.02
18 # Sample space - categories represent rarities
19 categories = [3, 4, 5, 6]
20
21
22
23 # Probability measure - assign probability based on pity,
24 # Value changes depending on the previous state
25 def probability_set(pity):
26     # Increase the probability of RaritySix
27     new_probability = [(((1 - pity) / 0.98) * 0.4), (((1 - pity)
28 ) / 0.98) * 0.5), (((1 - pity) / 0.98) * 0.08), pity]
29     # Decrease the other probabilities
30     probabilities = new_probability
31     # Normalization of the probabilities - to ensure their sum
32     equals to 1
33     total_probability = round(sum(new_probability), 16)
34     if total_probability != 1:
35         print(total_probability)
36         raise ValueError ("Sum of probabilities has to equal 1!
37         Instead its value is " + str(total_probability) + "!")
38     return probabilities
39
40
41
42 # Functions selecting specific operator in case defined rarity
43 has been selected
44 def select_operator_r3():
45     return ''.join(random.choices(rarity_three))
```

```
46
47 def select_operator_r4():
48     return ' '.join(random.choices(rarity_four))
49
50
51 def select_operator_r5():
52     # Specific operators rate up - 50% chance one of the 5 star
53     operators will be one of selected
54     rate_up = ['Franka', 'Proviso']
55     rate_up_choice = ['rate_up', 'no_rate_up']
56     chose = ' '.join(random.choices(rate_up_choice))
57     if chose == 'rate_up':
58         result = ' '.join(random.choices(rate_up))
59     elif chose == 'no_rate_up':
60         result = ' '.join(random.choices(rarity_five))
61     else:
62         raise ValueError ("Error, invalid value in rarity 5
63         selection.")
64     return result
65
66 def select_operator_r6():
67     rate_up = ['Mlynar']
68     rate_up_choice = ['rate_up', 'no_rate_up']
69     chose = ' '.join(random.choices(rate_up_choice))
70     if chose == 'rate_up':
71         result = ' '.join(random.choices(rate_up))
72     elif chose == 'no_rate_up':
73         result = ' '.join(random.choices(rarity_six))
74     else:
75         raise ValueError ("Error, invalid value in rarity 6
76         selection.")
77     return result
78
79 # Based on rarity this function selects operator
80 def select_operator(select_rarity):
81     selected = None
82     if select_rarity == 4:
83         selected = select_operator_r4()
84     elif select_rarity == 3:
85         selected = select_operator_r3()
```

```
86     elif select_rarity == 5:
87         selected = select_operator_r5()
88     elif select_rarity == 6:
89         selected = select_operator_r6()
90     return selected
91
92
93 # Function used to transfer data about each roll into csv
94 def csv_write_data(data):
95     with open('simulation.csv', 'a', newline='') as csvfile:
96         writer = csv.writer(csvfile)
97         writer.writerow(data)
98         print("_ _ _ _ _")
99         print(data)
100
101
102 #Function transfers data about results of simulation in a file
103 def csv_write_cost(rows_cost):
104     with open('cost.csv', 'a', newline='') as csvfile:
105         writer = csv.writer(csvfile)
106         writer.writerow(rows_cost)
107
108 # The main function of simulation
109 # number of rolls in single run defined by the variable "
    number_of_rolls"
110 def simulation():
111     # Define initial values
112     number_of_rolls = 1000
113     stop = False
114     roll = 0
115     pity = 0.02
116     change = False
117     pity_counter = 0
118     cost = 0
119     while not stop:
120         probabilities = probability_set(pity)
121         roll += 1
122         cost += 600
123         print("Pity counter: " + str(pity_counter)+"/50")
124         print(probabilities)
125         # Guaranteed rarity 5 or 6 in the first 10 rolls
126         if roll == 10 and change is False:
127             rarity = random.choices([5, 6], [0.8, 0.2])[0]
```

```
128         operator = select_operator(rarity)
129         data = [roll, rarity, operator]
130         csv_write_data(data)
131         if rarity != 6:
132             pity_counter += 1
133         else:
134             pity_counter = 0
135             pity = round(0.02, 2)
136             if operator == 'Mlynar':
137                 stop = True
138                 rows_cost = [roll, cost, 1]
139                 print('Simulation finished succesfully.')
140                 csv_write_cost(rows_cost)
141     elif roll <= number_of_rolls:
142         rarity = random.choices(categories, probabilities)
143     [0]
144         operator = select_operator(rarity)
145         data = [roll, rarity, operator]
146         # Write data in csv for further work with results
147         csv_write_data(data)
148         if rarity == 5:
149             change = True
150             # Uncomment to stop after rarity 5 obtained
151             #stop = True
152         if rarity != 6 and pity_counter < 50:
153             pity_counter += 1
154             if pity_counter == 50:
155                 pity += 0.02
156         elif rarity != 6 and pity_counter > 50:
157             pity += 1
158             pity = round(pity, 2)
159         elif rarity == 6:
160             # Uncomment to stop after rarity 6 obtained
161             #stop = True
162             change = True
163             pity = 0.02
164             pity_counter = 0
165             # Stop after target operator is obtained
166             if operator == 'Mlynar':
167                 stop = True
168                 rows_cost = [roll, cost, 1]
169                 print('Simulation finished succesfully.')
170                 csv_write_cost(rows_cost)
```

```
170         elif roll >= number_of_rolls:
171             roll -= 1
172             cost -= 600
173             stop = True
174             rows_cost = [roll, cost, 0]
175             csv_write_cost(rows_cost)
176             print('Simulation finished.')
177
178
179 # Run simulation 'n' number of times
180 # For result analysis only set 'n' to '0'
181 n = 10000
182 for _ in range(n):
183     simulation()
184     print('Simulation finished ' + str(n) + ' time(s).')
185
186 # DATA ANALYSIS PART
187 # Visualization of the outputs of simulation
188 df = pd.read_csv('simulation.csv')
189 print(df.describe() )
190
191 # Histogram
192 plt.hist(df['Rarity'], bins=4)
193 plt.title('Histogram of rarity distribution \n')
194 plt.xlabel('Rarity')
195 plt.ylabel('Number of rolls')
196 plt.show()
197
198 # Boxplot
199 df.boxplot('Roll number', by='Rarity')
200 plt.title('\n')
201 plt.xlabel('Rarity')
202 plt.ylabel('Roll number')
203 plt.show()
204
205 # Count number of occurrences of selected values in csv file
206 # Count number of occurrences of rarity six
207 df_result = df[df['Rarity'] == 6]
208 print(len(df_result))
209 # Count target operator occurrences
210 df_result = df[df['Operator'] == 'Mlynar']
211 print(len(df_result))
212
```

```
213 # Count number of occurrences of rarity five
214 df_result = df[df['Rarity'] == 5]
215 print(len(df_result))
216
217 # Data about results, cost prediction
218 df = pd.read_csv('cost.csv')
219 print(df.describe())
220
221 # Boxplot of simulation result
222 df.boxplot('Roll number', by='Result')
223 plt.title('\n')
224 plt.xlabel('Result')
225 plt.ylabel('Roll number')
226 plt.show()
```